INJECTIVITY OF A MODULE MAP OVER A VALUATION RING IS DETECTED ON THE SPECIAL FIBRE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 5043-5051 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 5053-5091, the complete author proof (37 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Valuation rings and their local homomorphisms, flatness criteria over a local ring, and essentially finitely type hypotheses; the reduction is retained.

Source reference. Original lemma 046Y: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 5091–5091. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0GSE: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 5091–5091. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let A be a valuation ring. Let $A \to B$ is a local homomorphism of local rings which is essentially of finite type. Let $u : N \to M$ be a map of finite B-modules. Assume M is flat over A and $\overline{u} : N/\mathfrak{m}_A N \to M/\mathfrak{m}_A M$ is injective. Then u is injective and $M/u(N)$ is flat over A.*

Proof. We will deduce this lemma from Algebra, Lemma 046Y (please note that we exchanged the roles of M and N). To do the reduction we will use More on Algebra, Lemma 0GSE to reduce to the finitely presented case.

By assumption we can write B as a quotient of the localization of a polynomial algebra $P = A[x_1, \ldots, x_n]$ at a prime ideal $\mathfrak{q}$. Then we can think of $u : N \to M$ as a map of finite $P_\mathfrak{q}$-modules. Hence we may and do assume that B is essentially of finite presentation over A.

Next, the B-module N is finite but perhaps not of finite presentation. Write $N = \operatorname{colim} N_\lambda$ as a filtered colimit of finitely presented B-modules with surjective transition maps. For example choose a presentation $0 \to K \to B^{\oplus r} \to N \to 0$, write K as the union of its finite submodules K_λ, and set $N_\lambda = \operatorname{Coker}(K_\lambda \to B^{\oplus r})$.

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The module $N/\mathfrak{m}_A N$ is of finite presentation over the Noetherian ring $B/\mathfrak{m}_A B$. Hence for λ large enough we have $N_\lambda/\mathfrak{m}_A N_\lambda = N/\mathfrak{m}_A N$. Now, if we can show the lemma for the composition $u_\lambda : N_\lambda \to M$, then we conclude that $N_\lambda = N$ and the result holds for u. Hence we may and do assume N is of finite presentation over B.

By More on Algebra, Lemma 0GSE the module M is of finite presentation over B. Thus all the assumptions of Algebra, Lemma 046Y hold and we conclude. $\square$